Kodewala Academy (25th Aug, 2026) - Java Backend Test # 5

Student: AnuragRamtekkar (Online) | Oct. 29, 2026, 7:11 p.m. | Duration: 0:28:05

Score: 76% (31 / 41) **Correct: 33** **Incorrect: 8 (-0.25 each)**

Green = correct answer Red = your wrong answer (a tick marks your selection)

1. What is the output? [1 / 1 point]

```
String s1 = new String("Java");
String s2 = new String("Java");
System.out.println(s1 == s2);
```

[ ] A. true
[x] B. false (correct)
[ ] C. 0
[ ] D. 1

2. What is the output? [1 / 1 point]

```
String s1 = "Java";
String s2 = new String("java");
System.out.println(s1.equals(s2));
```

[ ] A. true
[ ] B. Compilation Error
[x] C. false (correct)
[ ] D. 1

3. What is the output? [1 / 1 point]

```
String s = "Hello";
s.concat(" World");
System.out.println(s);
```

[ ] A. Hello World
[ ] B. Compilation Error
[ ] C. Runtime Error
[x] D. Hello (correct)

4. Which method compares String contents? [1 / 1 point]

[ ] A. ==
[x] B. equals() (correct)
[ ] C. hashCode()
[ ] D. compare()

5. Which statement is TRUE? [1 / 1 point]

[ ] A. String is mutable
[ ] B. StringBuilder is immutable
[x] C. String is immutable (correct)
[ ] D. StringBuffer is immutable

6. Which method is inherited by every Java class? [1 / 1 point]

[ ] A. None
[x] B. Both - equals() and hashCode() (correct)
[ ] C. hashCode()
[ ] D. equals()

7. What happens if equals() is not overridden? [-0.25 / 1 point]

[x] A. Compares content (your answer)
[ ] B. Uses == internally (correct)
[ ] C. Compilation Error
[ ] D. Runtime Exception

8. Which modifier is generally used for encapsulation (to hide data)? [1 / 1 point]

[ ] A. public
[ ] B. protected
[x] C. private (correct)
[ ] D. default

9. Getter methods should generally be [1 / 1 point]

[x] A. public (correct)
[ ] B. protected
[ ] C. private
[ ] D. default

10. final variable [1 / 1 point]

[x] A. Cannot change reference/value after assignment (correct)
[ ] B. Cannot be initialized
[ ] C. Can be reassigned
[ ] D. None

11. final method [1 / 1 point]

[ ] A. Cannot be overloaded
[x] B. Cannot be overridden (correct)
[ ] C. Cannot be called
[ ] D. Cannot return value

12. final class [1 / 1 point]

[ ] A. Cannot create object
[ ] B. Must be abstract
[x] C. Cannot be inherited (correct)
[ ] D. Can be inherited

13. Which class is final? [-0.25 / 1 point]

[x] A. Object (your answer)
[ ] B. Thread
[ ] C. String (correct)
[ ] D. Exception

14. When is constructor called? [1 / 1 point]

[ ] A. Method call
[x] B. Object creation (correct)
[ ] C. Compilation
[ ] D. Garbage collection

15. If no constructor is written [1 / 1 point]

[ ] A. Compilation Error
[ ] B. JVM creates
[x] C. Compiler provides default constructor (correct)
[ ] D. None

16. First statement inside constructor can be [1 / 1 point]

[ ] A. super()
[ ] B. this()
[x] C. either super() or this() (correct)
[ ] D. anything

17. Which executes first? [1 / 1 point]

[ ] A. Constructor
[ ] B. IIB
[ ] C. main method
[x] D. SIB (correct)

18. IIB executes [1 / 1 point]

[ ] A. Once per class
[x] B. Every object creation (correct)
[ ] C. Before main()
[ ] D. After main()

19. Execution order [1 / 1 point]

[ ] A. Constructor -> IIB -> SIB
[x] B. SIB -> IIB -> Constructor (correct)
[ ] C. IB -> Constructor -> SIB
[ ] D. Constructor -> SIB -> IIB

20. Which keyword invokes parent constructor? [1 / 1 point]

[ ] A. parent()
[ ] B. this()
[x] C. super() (correct)
[ ] D. extends()

21. Object class is [1 / 1 point]

[x] A. Parent of all classes (correct)
[ ] B. Interface
[ ] C. Final class
[ ] D. Abstract class

22. Instance variable is created [1 / 1 point]

[ ] A. Once

[ ] B. During class loading

[x] C. With every object (correct)

[ ] D. During compilation

23. A Java application receives thousands of requests simultaneously. Each request builds a large string by appending multiple values. The same string object is shared by multiple threads. [-0.25 / 1 point]

[ ] A. String

[x] B. StringBuilder (your answer)

[ ] C. StringBuffer (correct)

[ ] D. Thread

24. Why is StringBuilder preferred over StringBuffer in this program? [1 / 1 point]

```
public class Demo {
    public static void main(String[] args) {
        StringBuilder sb = new StringBuilder();

        for (int i = 0; i < 100000; i++) {
            sb.append(i);
        }

        System.out.println(sb.length());
    }
}
```

[ ] A. StringBuilder uses less memory

[ ] B. StringBuilder is immutable.

[ ] C. StringBuffer cannot append integers.

[x] D. The program is single-threaded, so synchronization overhead is unnecessary. (correct)

25. A developer accidentally loses the source code (.java files), but the compiled .class files are still available. Which statement is correct? [1 / 1 point]

[ ] A. It is impossible to recover the Java code.

[x] B. A Java decompiler can generate readable Java source code from the .class files, although comments and original formatting are usually lost. (correct)

[ ] C. The JVM can automatically recreate the original source code with comments.

[ ] D. The .class file contains only machine code, so decompilation is impossible.

26. Employee does NOT override equals(). What happens? [1 / 1 point]

```
Employee e1 = new Employee(101);
Employee e2 = new Employee(101);
System.out.println(e1.equals(e2));
```

[ ] A. true

[x] B. false (correct)

[ ] C. Compilation Error

[ ] D. Runtime Exception

27. A banking application updates a shared transaction log using multiple threads simultaneously. Which class should be used? [1 / 1 point]

[ ] A. String
[ ] B. StringBuilder
[x] C. StringBuffer (correct)
[ ] D. char[]

28. During testing, some modules assign negative marks. What is the best design improvement? [1 / 1 point]

```
class Student {
    public int marks;
}
```

[ ] A. Make marks final.
[x] B. Make marks private and validate in setter. (correct)
[ ] C. Make marks protected.
[ ] D. char[]

29. An e-commerce application caches products using product IDs as String keys. What will be printed? [1 / 1 point]

```
String id1 = new String("P101");
String id2 = new String("P101");
if(id1 == id2){
    System.out.println("Cache Hit");
}else{
    System.out.println("Cache Miss");
}
```

[ ] A. Cache Hit
[x] B. Cache Miss (correct)
[ ] C. Compilation Error
[ ] D. Depends on JVM

30. How many String objects are created at runtime? [-0.25 / 1 point]

```
String s = "Java" + "Backend";
```

[x] A. 0 (your answer)
[ ] B. 1 (correct)
[ ] C. 2
[ ] D. 3

31. An application loads configuration only once when the application starts. Which Java feature is most suitable for performing this initialization? (Java version is 17) [1 / 1 point]

[ ] A. Constructor
[ ] B. Instance Initialization Block (IIB)
[x] C. Static Initialization Block (SIB) (correct)
[ ] D. finalize()

32. A Product object is created for every incoming request. Before every constructor execution, default values must be initialized. Which feature is most appropriate? [1 / 1 point]

[ ] A. Static block

[x] B. Instance Initialization Block (IIB) (correct)

[ ] C. finalize()

[ ] D. Static method

33. Why is the constructor made private? [1 / 1 point]

```
public class Utility {
    private Utility()
    {
    }
}
```

[ ] A. To improve performance.

[x] B. To prevent object creation because the class contains only utility/static methods. (correct)

[ ] C. To support inheritance.

[ ] D. To reduce memory usage.

34. You are working on a Java-based Order Management System. During testing, the application prints "New Order". However, the business requirement is that two orders with the same orderId should be considered equal. What is the best solution? [1 / 1 point]

```
public class Order {
    private String orderId;
    public Order(String orderId)
    {
        this.orderId = orderId;
    }
}

Order o1 = new Order("ORD101");
Order o2 = new Order("ORD101");
if (o1.equals(o2)) {
    System.out.println("Duplicate Order");
} else {
    System.out.println("New Order");
}
```

[ ] A. Compare o1 == o2

[ ] B. Override only hashCode().

[x] C. Override both equals() and hashCode() based on orderId. (correct)

[ ] D. Make orderId static.

35. What is the output? [-0.25 / 1 point]

```
class PaymentService {
    static void process() {
        System.out.println("PaymentService");
    }
}
class UpiPaymentService extends PaymentService {
    static void process() {
        System.out.println("UpiPaymentService");
    }
}
public class PaymentApplication {
    public static void main(String[] args) {
        PaymentService payment = new UpiPaymentService();
        payment.process();
    }
}
```

[ ] A. PaymentService (correct)

[x] B. UpiPaymentService (your answer)

[ ] C. Compilation error

[ ] D. Runtime exception

36. What is the output? [-0.25 / 1 point]

```
class NotificationService {
    NotificationService() {
        sendNotification();
    }
    void sendNotification() {
        System.out.println("Default Notification");
    }
}
class EmailNotificationService extends NotificationService {
    String email = "admin@company.com";
    @Override
    void sendNotification() {
        System.out.println(email);
    }
}
public class NotificationApplication {
    public static void main(String[] args) {
        new EmailNotificationService();
    }
}
```

[ ] A. admin@company.com

[ ] B. null (correct)

[ ] C. Default Notification

[x] D. Compilation error (your answer)

37. What is the output? [-0.25 / 1 point]

```
class OrderProcessor {
    void process(long orderId) {
        System.out.println("Long");
    }

    void process(Integer orderId) {
        System.out.println("Integer");
    }
}

public class OrderApplication {
    public static void main(String[] args) {
        OrderProcessor processor = new OrderProcessor();
        processor.process(100);
    }
}
```

[ ] A. Long (correct)

[x] B. Integer (your answer)

[ ] C. Compilation error

[ ] D. Ambiguous method call

38. What is the output? [-0.25 / 1 point]

```
class Employee {
    String role = "Employee";
    String getRole() {
        return role;
    }
}
class SoftwareEngineer extends Employee {
    String role = "Software Engineer";
    @Override
    String getRole() {
        return role;
    }
}
public class HRApplication {
    public static void main(String[] args) {
        Employee employee = new SoftwareEngineer();
        System.out.println(employee.role);
        System.out.println(employee.getRole());
    }
}
```

[ ] A. Employee / Employee

[x] B. Software Engineer / Software Engineer (your answer)

[ ] C. Employee / Software Engineer (correct)

[ ] D. Software Engineer / Employee

39. An e-commerce application has an OrderService class that performs three tasks: calculates order totals, saves orders to MySQL, and sends confirmation emails. What is the best design improvement according to SRP? [1 / 1 point]

[x] A. Create separate classes: OrderCalculator, OrderRepository, and EmailService. (correct)

[ ] B. Create three subclasses of OrderService, one for each task.

[ ] C. Keep all methods in OrderService but make them private.

[ ] D. Convert OrderService into an abstract class and move the email logic to a subclass.

40. A payment application supports credit card and UPI payments. Whenever a new payment method is introduced, developers must modify this existing class. Which design best follows OCP? [1 / 1 point]

```
class PaymentService {
    void pay(String type, double amount) {
        if (type.equals("CARD")) {
            // Process card payment
        } else if (type.equals("UPI")) {
            // Process UPI payment
        }
    }
}
```

[ ] A. Add another else if block for every new payment method.

[x] B. Create a Payment interface with a pay() method and implement it in CardPayment, UpiPayment, and other payment classes. (correct)

[ ] C. Make PaymentService a singleton.

[ ] D. Use static methods for each payment type inside PaymentService.

41. In Java 7, which of the following statements about interfaces is correct? [1 / 1 point]

[ ] A. An interface can contain default methods with implementation.

[ ] B. An interface can contain static methods with implementation.

[x] C. An interface can contain public static final variables and abstract methods. (correct)

[ ] D. An interface can contain a constructor to initialize its variables.